

ABSTRACT

The MyDiagnexia Quality Plan describes how Diagnexia will launch and operate a **safe, compliant, and scalable web-based digital pathology platform for the U.S. market**. It focuses on three things: (1) keeping patients safe, (2) meeting CLIA/CAP/ISO expectations, and (3) ensuring the product is release-ready and auditable at every stage of its lifecycle.

At a high level, this plan explains **what MyDiagnexia does, which rules it must follow, and how the company will prove it is in control**. MyDiagnexia provides a cloud-based digital pathology workflow that includes slide upload and QC, a zero-footprint web viewer, web-based reporting, case assignment, LIS/LIMS integrations, and support for pathologists using their own devices (BYOD). It does **not** perform automated diagnosis and is **not** regulated as an FDA medical device; instead, it sits under CLIA, CAP, ISO 15189 and Diagnexia's Integrated Management System (IMS).

The Quality Plan aligns these features with Diagnexia's **core quality system**: document control, risk management, secure SDLC, computerized system validation (CSV), change control, incident/CAPA management, training, and security/privacy controls. In practice, that means every release of MyDiagnexia is backed by a **validated build, documented risk assessment, controlled change record (ECO), and training for the people who use or support it**. All critical records live in Orcanos to ensure full traceability.

Operationally, the plan defines **who is accountable for what**. Product and Engineering own requirements and implementation; QA owns validation strategy, documentation control and audit readiness; Regulatory ensures CLIA/CAP and state rules are met; Information Security owns HIPAA and ISO 27001 controls; Clinical Operations and Customer Success own deployment and customer validation; the U.S. Medical Director is the final clinical authority approving validation and patient use. These owners are named explicitly in the Roles & Responsibilities section and reflected again in the activity-level accountability tables throughout the document.

Finally, the plan sets expectations for **ongoing control after go-live**. Post-implementation monitoring will track viewer performance, upload success, QC defects, integration failures and turnaround time, with issues routed through Incident Management and CAPA where needed. Retirement of any component requires a controlled change, data retention in line with CLIA/CAP expectations, and secure decommissioning. Taken together, this Quality Plan provides a single, end-to-end view of how Diagnexia will safely move MyDiagnexia from development to clinical use, and then keep it in a validated, compliant state over time.

1. PURPOSE

This Quality Plan describes the **quality framework, lifecycle activities, validation strategy, roles, controls, documentation requirements, and regulatory compliance obligations** governing the development, deployment, maintenance, and continuous improvement of the **MyDiagnexia Platform**.

It ensures that all MyDiagnexia releases are:

- Fit for their intended use
- Compliant with CLIA, CAP, ISO 15189 and Diagnexia's Integrated Management System (IMS)
- Validated according to GAMP5 and Diagnexia SSDLC (DMS-8333)

- Implemented with adequate risk control, training, documentation, and change management
- Safe, secure, and reliable for clinical operations in the USA

This Quality Plan covers all phases (Phase 1, Phase 2, Phase 3 and future modules), and all functionality within the MyDiagnexia ecosystem.

2.1 Systems in Scope

This Quality Plan applies to all U.S. deployments of the MyDiagnexia platform, covering the full set of functional components, supporting modules, and integrations required to deliver a compliant digital pathology workflow. The scope includes the core elements of the system such as the MyDiagnexia Zero-Footprint (ZFP) Web Viewer, the QC Cloud module responsible for image quality control and upload workflows, and the Hubless Upload mechanism used to transmit scanner-produced digital images directly into the cloud environment. It also encompasses the Pathologist Portal, Referring Lab Manager Portal, slide manifest and metadata management processes, and the manual and rules-based Pathologist Case Assignment Engine. Additional core components include cloud storage and image rendering capabilities, account management functions (including SSO where applicable), the client onboarding modules, BYOD workflows that support pathologist-owned devices, historical case retrieval functions, and billing metadata management for professional-component (PC-only) use cases.

The scope further extends to Phase 2 system components, which introduce enhanced reporting and productivity features. These include Editable Web Reporting, pathologist dashboards, consult workflows, measurement tools, CPT/ICD metadata population, and non-diagnostic AI safeguards used to enhance reporting safety and minimize human-factor errors.

Additionally, this Quality Plan includes Phase 3 functionality, which expands the platform into credentialing and resource-management domains. These modules comprise the pathologist credentialing and onboarding workflows, a pathologist profile and configuration module, auto-assignment logic for case routing, case load and capacity controls, and supporting APIs including integration with BambooHR for workforce management.

Lastly, the Quality Plan also covers all technical integrations required for MyDiagnexia operations, including LIS/LIMS connectivity via HL7 (v2.x) and FHIR, scanner integrations for devices such as Hamamatsu, Leica, Roche, Glissando, and DICOM conversion tools, U.S.-based AWS cloud deployment, and future integrations such as Coronis revenue cycle management (RCM) connectivity. Cross-border collaborations, such as interaction with Alpha Labs Canada where applicable, are also included to ensure consistent quality and regulatory alignment.

2.2 Regulatory Scope

The regulatory scope of this Quality Plan includes all standards, laws, and accreditation requirements that govern the design, deployment, validation, and clinical use of the MyDiagnexia platform. As a non-FDA-regulated digital workflow system that does not perform automated diagnosis and relies on Research Use Only (RUO) scanners, MyDiagnexia falls under a framework focused on clinical laboratory quality, data integrity, and patient safety, rather than medical device regulation. Accordingly,

this Quality Plan ensures full alignment with CLIA regulations (42 CFR 493), which mandate the quality, accuracy, and reliability of laboratory testing processes, including digital pathology workflows, result reporting, personnel competency, and system integrity. It likewise incorporates all applicable CAP accreditation checklist requirements, particularly those from the GEN, ANP, COM, TRM, QMS, and LIS sections that address digital slides, image management, reporting, QC processes, informatics, and laboratory information system validation.

This plan also aligns with ISO 15189:2022, as reflected in Diagnexia's Quality Manual (**DMS-8099**), ensuring that MyDiagnexia supports the broader quality and competence framework required for medical laboratory operations. Furthermore, because the platform processes protected health information, it adheres to HIPAA, relevant U.S. state privacy requirements, and contractual privacy and security obligations. The platform's cloud infrastructure, security controls, and information governance processes are also developed in accordance with ISO 27001, which forms part of Diagnexia's Integrated Management System (IMS) and ensures structured, auditable, and comprehensive information security management.

In addition to these primary regulatory obligations, the Quality Plan leverages GAMP5 principles to define and control the system development lifecycle (SDLC), validation activities, testing rigor, and documentation requirements appropriate for a complex Category 4–5 software platform. Collectively, these frameworks ensure that MyDiagnexia is validated, deployed, and maintained in a manner consistent with the operational, ethical, and regulatory expectations for clinical laboratories, while explicitly excluding FDA Quality System Regulation and medical device controls, which do not apply to RUO devices or non-SaMD workflow platforms.

3. ROLES & RESPONSIBILITIES

Role	Responsibilities
CIO / IT Lead	Ensure infrastructure qualification, cloud security, U.S. region compliance.
Product Owner	Own URS, scope, prioritization, acceptance criteria.
Engineering / Development Lead	Develop software per SSDLC (DMS-8333), coding standards (DMS-SOP-18).
QA (USA)	Author this Quality Plan; ensure validation, documentation, traceability, audit readiness.
Regulatory Affairs (USA)	Ensure CLIA, CAP, state licensure compliance; oversee HIPAA and privacy alignment.

U.S. Medical Director (CLIA Director)	Validate clinical workflows, approve PQ, approve production release.
Security Officer	Ensure ISO 27001 compliance and incident response.
Customer Success / Implementation	Execute customer validation, onboarding steps, training delivery.
Pathologist End Users	Validate BYOD devices, complete training, perform UAT workflows.

3.1 Accountability Overview

The table below summarizes who is accountable for each major quality activity within the MyDiagnexia program. Detailed RACI-level expectations are embedded in each section of this plan.

Area / Activity	Primary Owner (A)	Supporting Functions
Overall Quality Plan & audit readiness	QA (USA)	Regulatory, Engineering, Clinical Ops
Requirements & scope (URS, roadmap)	Product Owner	Engineering, QA, Regulatory, Medical Dir.
Secure SDLC & development	Engineering / Dev Lead	Product, QA, InfoSec
Computerized system validation (IQ/OQ/PQ)	QA	Engineering, Clinical Ops, Medical Director
Clinical approval of validation	U.S. Medical Director	QA, Regulatory
Risk management (QRA, DPIA)	QA (USA)	Regulatory, InfoSec, Product
Change control / ECOs	QA (USA)	Product, Engineering, Regulatory, IS, DPO
Incident & CAPA	QA	InfoSec, Engineering, Clinical Ops, DPO
Training & competency	QA + Training Lead	HR/BambooHR, Clinical Ops, Product
Security & privacy controls	Information Security	QA, DPO, Engineering
LIS/LIMS integrations	Integration Engineering	QA, Customer Success
Customer validation (UAT)	Customer Success / Impl.	QA, Integration, Clinical Ops
Release management	QA + Engineering	Product, Regulatory, InfoSec,

		Medical Dir.
Post-implementation monitoring	QA + Clinical Ops	Engineering, Customer Success

QA Responsibilities Summary: QA (USA) provides life-cycle oversight for validation, documentation, change control, risk management, training compliance, and audit readiness across all MyDiagnexia phases.

4. SYSTEM OVERVIEW

MyDiagnexia is a cloud-based digital pathology workflow system that enables:

- Digital slide ingestion (cloud-based)
- QC and metadata validation
- Pathologist routing, assignment, and reporting
- Secure cloud-rendered image viewing (ZFP viewer)
- Multi-site and multi-pathologist collaboration
- Integration with external LIS systems
- Non-diagnostic AI for QC and reporting safeguards

MyDiagnexia does not perform automated diagnosis and does not constitute an FDA-regulated medical device.

5. QUALITY MANAGEMENT FRAMEWORK

The MyDiagnexia platform operates under the governance of Diagnexia's Integrated Management System (IMS), which unifies the requirements of ISO 15189, ISO 27001, CLIA, and CAP into a coherent and auditable quality management structure. This Quality Plan is built upon the principles, processes, and expectations established in the Diagnexia Quality Manual (DMS-8099), ensuring that all activities related to system development, validation, deployment, and ongoing operation are handled in a manner consistent with internationally recognized laboratory and information security standards.

The framework relies heavily on Diagnexia's controlled suite of quality procedures, including the Document Control Procedure (DMS-8338) and the Record Management Procedure (DMS-8079), which govern document creation, approval, distribution, retention, and archival activities to maintain accuracy and integrity throughout the product lifecycle. Risk identification, assessment, and mitigation are executed under the Quality, Governance and Risk Procedure (DMS-8035), ensuring that risks

related to digital workflows, clinical impact, cyber threats, or operational failures are appropriately documented, monitored, and addressed.

Software development, enhancement, and maintenance activities follow the Secure Software Development Life Cycle (SSDLC) requirements outlined in DMS-8333, which incorporate GAMP5 principles and ISO 27001-aligned security controls to ensure traceability, secure coding, peer review, testing rigor, and auditability across all phases of development. All computerized system validation (CSV) activities—including testing, qualification, deviations, and approvals—are executed in alignment with DMS-8085, which defines the expectations for Installation Qualification (IQ), Operational Qualification (OQ), Performance Qualification (PQ), and validation documentation. Training and competency requirements are governed by the Training Procedure (DMS-8082), ensuring that all users, developers, support staff, and clinical personnel are adequately trained and qualified.

Operational issues, system problems, or deviations are documented and remediated using the Incident Management Procedure (DMS-8077) and the Corrective and Preventive Action (CAPA) process contained in DMS-8295, ensuring that any identified nonconformities are resolved in a compliant and traceable manner. Configuration and change management are strictly governed by DCX-SOP-28, ensuring that each software change or configuration modification undergoes an impact assessment, controlled approval, and appropriate testing before deployment into any validated environment.

Collectively, this Quality Management Framework ensures that MyDiagnexia is developed, validated, released, and maintained in a state of control throughout its entire lifecycle, supporting compliance with CLIA and CAP standards and safeguarding patient data, system integrity, and clinical reliability.

6. USER REQUIREMENTS SPECIFICATION (URS)

A full URS shall be authored and approved before validation.

Modules include:

- Viewer requirements
- Web reporting requirements
- QC Cloud requirements
- Upload & metadata requirements
- Admin portal requirements
- LIS integration requirements
- BYOD device requirements
- Security and privacy requirements

The User Requirements Specification (URS) defines the functional, technical, operational, and regulatory expectations for the MyDiagnexia platform and serves as the foundational document guiding all subsequent design, development, and validation activities. The URS captures the needs of pathologists, laboratory personnel, administrators, integration partners, and quality and regulatory

stakeholders to ensure that the system supports safe, compliant, efficient, and reliable digital pathology workflows across all phases of deployment. It articulates requirements for core platform functionalities—including image upload, quality control, metadata handling, case assignment, and cloud rendering—as well as advanced modules such as web-based reporting, pathologist dashboards, credentialing workflows, administrative features, and interoperability with external systems.

In alignment with Diagnexia's Secure Software Development Life Cycle (SSDLC; DMS-8333), the URS establishes requirements for system performance, usability, security, auditability, privacy, data integrity, and scalability. It incorporates CLIA and CAP expectations related to digital pathology operations, including traceability, turnaround time, reporting accuracy, and integration with laboratory information systems (LIS/LIMS). The URS also includes specifications for browser compatibility, pathologist device requirements, account controls, user interface behavior, measurement tools, workflow logic, and non-diagnostic AI safeguards used to prevent reporting inconsistencies or user errors.

The URS must be reviewed and approved by Product Management, Engineering, QA, Regulatory Affairs, and the U.S. Medical Director to ensure that system requirements align with the intended use defined in this Quality Plan. Each URS requirement will be linked to corresponding design elements, test protocols, and validation evidence through the Requirement Traceability Matrix (RTM), ensuring end-to-end coverage during verification and validation activities.

7. RISK MANAGEMENT

The purpose of risk management in MyDiagnexia is to proactively identify where things can go wrong and to put controls in place before patients, data, or operations are harmed. Risk management for the MyDiagnexia platform is conducted in accordance with Diagnexia's Quality, Governance and Risk Procedure (DMS-8035) and follows the structured Quality Risk Management (QRM) framework outlined in Diagnexia's Integrated Management System (IMS). Given that MyDiagnexia introduces decentralized digital pathology workflows, client-side uploads, cloud-based reporting modules, and the use of external pathologist devices, a comprehensive risk management process is essential to ensure that patient safety, data integrity, and regulatory compliance are maintained throughout the system's lifecycle.

A formal Quality Risk Assessment (QRA) has been completed for the MyDiagnexia program (DMS-105073), identifying key risk domains spanning regulatory compliance, HIPAA-aligned privacy practices, international privacy frameworks, information security, scanner interoperability, AI-supported QC safeguards, operational readiness, business continuity, and long-term data retention. The QRA highlights significant risks associated with interstate and international case routing, including potential implications under CLIA, CAP, and malpractice standards if case ownership, reporting identifiers, or laboratory director responsibilities are not clearly defined. Risks related to PHI access across decentralized devices—including BYOD laptops used by pathologists—were classified as *extreme*, requiring enforcement of access controls (SSO/MFA), minimum device security specifications, full-disk encryption, and updated Data Protection Impact Assessments (DPIAs).

Additional risks were identified in technical areas such as OCR/case ID mismatches, scanner interoperability across TIFF/SVS/NDPI/DICOM formats, metadata errors, usability gaps in web-based reporting, and variability in pathologist device performance. To mitigate these issues, the QRA calls for enhancements to QC workflows, DICOM validation pipelines, fallback mechanisms for metadata

correction, and formal usability and performance testing of reporting modules and screen-size minimums before release. Risks related to AI functionality—specifically AI-driven QC and non-diagnostic reporting safeguards—require validation prior to any clinical application and must remain strictly outside the scope of automated clinical interpretation to avoid triggering FDA oversight.

Operational risks also feature prominently, including gaps in training and user onboarding, regulatory nonconformance due to incomplete SOP updates, misalignment in billing metadata, and inadequate workflow documentation across global regions. Corrective actions outlined in the QRA include updates to user guides, structured instructor-led training with attestation, validation of storage and retention systems such as WASABI to meet CAP/CLIA timeframes, and reinforcement of disaster recovery procedures consistent with DMS-8080 and DMS-72576.A1 (Master Security Document) .

The risk management approach ensures that each identified risk is evaluated for severity, likelihood, and impact using Diagnexia's standard scoring criteria, with mitigation strategies and corrective actions tracked in a structured CAPA process. Dependencies across multiple ECOs—including QC tooling, scanner integration, case assignment logic, and AI model updates—are reviewed collectively to avoid introducing unintended risks due to overlapping releases or configuration changes .

Residual risk for MyDiagnexia is considered acceptable only upon completion of corrective actions such as DPIA updates, scanner interoperability validation, AI safeguard verification, SOP updates, pathologist training records, global LIS testing alignment, and confirmation of business continuity readiness. The QRA and its associated risk files will be reviewed periodically—at minimum quarterly post-launch—and updated as mitigation actions are implemented or new risks emerge during system scaling.

8. TRACEABILITY

Traceability for the MyDiagnexia platform is maintained through a structured, end-to-end framework that links user requirements, functional designs, risk controls, testing activities, and validation evidence across the full lifecycle of the system. In alignment with Diagnexia's Secure Software Development Life Cycle (SSDLC; DMS-8333) and Computerized System Validation Procedure (DMS-8085), all requirements captured in the User Requirements Specification (URS) are mapped to corresponding functional specifications, design elements, test cases, and qualification results within the Requirement Traceability Matrix (RTM). This ensures that every feature—whether part of digital slide upload, QC Cloud, viewer rendering, web reporting, LIS integration, BYOD workflows, pathologist assignment logic, or administrative functions—can be directly traced from its original requirement to its verification and final approval.

Design reviews occur at multiple lifecycle stages to confirm that functional specifications are complete, technically feasible, and aligned with clinical and regulatory expectations. These reviews include evaluation of architecture diagrams, data flow models, security configurations, metadata schemas, workflow logic, and scanner interoperability requirements, including DICOM compatibility as identified in the Quality Risk Assessment (DMS-105073) . Any risks identified during design review—such as metadata mismatch potential, regional reporting requirements, cross-border routing constraints, or browser/device variability—are incorporated into the Functional Risk Assessment (FRA) and linked to specific mitigation actions or test cases as appropriate.

The RTM plays a central role during validation, establishing transparent traceability across IQ, OQ, PQ, and any required PPQ or customer validation steps. This ensures that each requirement has objective evidence of verification and that gaps, deviations, or changes introduced through

Engineering Change Orders (ECOs) are properly controlled under DCX-SOP-28. As multiple technical workstreams evolve in parallel—including QC tooling updates, scanner integration enhancements, cloud storage configuration, reporting interface refinements, and credentialing workflow development—the traceability structure prevents loss of control and supports regulatory audit readiness by clearly demonstrating how each feature was validated, assessed for risk, and approved for use.

Comprehensive traceability also supports ongoing regulatory obligations, including CAP requirements for telepathology validation (GEN.40630, GEN.50630, GEN.52920) and CLIA expectations related to LIS testing, software changes, and system verification. By maintaining continuous alignment between requirements, risks, controls, and validation outcomes, the traceability process ensures that MyDiagnexia remains compliant, internally consistent, and fully defensible across all development cycles and system enhancements.

RTM will map:

- **URS → FSDS → Test Cases → PQ → Release Criteria**

Document: DCX-RTM-MyDiagnexia

9. SOFTWARE DEVELOPMENT METHODS

Software development must comply with:

- **DMS-8333 Secure SDLC**
- **DMS-SOP-18 Programming Standards**
- **Agile methodology for feature releases**
- **Waterfall methodology for validation packages**
- **GitLab version control**
- **Peer code review requirements**

The development of the MyDiagnexia platform is governed by Diagnexia's Secure Software Development Life Cycle (SSDLC) Policy (DMS-8333), which establishes structured, traceable, and risk-based controls for all software engineering activities. This lifecycle framework applies to every component of the platform—including the Zero-Footprint Web Viewer, QC Cloud, case assignment logic, Web Reporting modules, scanner interoperability layers, LIS integrations, and administrative functions—and ensures that software is built, tested, secured, and released in accordance with recognized industry standards and the regulatory expectations applicable to digital pathology workflows.

MyDiagnexia follows an agile development model for iterative feature implementation while adhering to a more formal, GAMP5-aligned process for system qualification and release into validated environments. Requirements captured in the URS are decomposed into functional and design

specifications, which inform development tasks, user stories, and acceptance criteria. All coding activity must comply with Diagnexia's Programming Principles and Standards (DMS-SOP-18), including conventions for secure coding, documentation, modularity, and peer review. Code changes undergo mandatory peer review, vulnerability checks, and automated testing through the CI/CD pipeline prior to merging into controlled branches.

Design verification and feature-level testing are performed throughout development, including unit tests, integration tests, API validation, browser compatibility testing, performance checks, and interoperability validation—particularly for scanner file formats and DICOM conversion workflows, which were identified as key risk areas in the Quality Risk Assessment (DMS-105073) . Any user-facing module, such as reporting, case routing, viewer interactions, or device-dependent workflows (e.g., BYOD), must undergo usability and functional review to ensure alignment with clinical expectations, privacy obligations, and system safety parameters.

Changes introduced through Engineering Change Orders (ECOs)—including enhancements to QC tooling, modifications to case assignment logic, cloud storage configuration updates, and integration features—are controlled under the Configuration & Change Management Procedure (DCX-SOP-28). This process mandates impact assessments, regression testing, risk evaluation, documentation revision, and approval prior to promotion into validation or production environments.

Collectively, these software development methods ensure that MyDiagnexia is engineered using secure, traceable, and quality-controlled processes that meet Diagnexia's internal standards, support CAP and CLIA compliance, and ensure that each release is robust, auditable, and fit for its intended use in clinical digital pathology workflows.

10. VALIDATION STRATEGY

The purpose of the validation strategy is to prove that MyDiagnexia works as intended in real clinical workflows before we use it on patient cases. The validation strategy for the MyDiagnexia platform is built upon Diagnexia's established computerized system validation (CSV) framework and adheres to the regulatory expectations defined by CAP, CLIA, and ISO 15189. Validation activities are conducted in accordance with the Diagnexia US Clinical Validation / Verification Plan (DMS-108833), which outlines the principles, workflow requirements, documentation structure, and approval hierarchy required to qualify digital pathology systems, telepathology workflows, and operational processes prior to clinical use . The strategy ensures that each system enhancement, workflow change, integration update, or feature release is appropriately validated to confirm that the platform continues to support accurate, secure, and reliable clinical diagnostic workflows within a CLIA-certified and CAP-accredited environment.

Validation encompasses multiple layers of testing, starting with installation qualification (IQ), followed by operational qualification (OQ), and concluding with performance qualification (PQ). IQ verifies that the system is deployed in a controlled, documented manner within the U.S. validation environment, including confirmation of the build ID, URL, workstation configuration, hardware specifications, and security controls. OQ testing confirms that each software feature operates according to design specifications and acceptance criteria defined in applicable requirements documents, including test cases derived from the URS, Functional Requirements Specifications (FRS), and identified risks. PQ testing evaluates system performance under intended-use conditions, reflecting real clinical workflows such as digital slide ingestion, QC Cloud assessment, viewer performance, reporting actions, integration behavior, and case routing logic. As described in DMS-108833, PQ testing must emulate

authentic pre-analytic, analytic, and post-analytic processes, and must meet CLIA performance parameters for accuracy, precision, analytical sensitivity/specificity, and reportable range when applicable .

Risk-based validation principles guide the selection of what must be tested for each release. The Clinical Validation Plan requires that a project-level Quality Risk Assessment (DMS-105073) be completed to identify high-risk changes requiring testing versus low-risk changes that may be sufficiently supported through peer-reviewed literature, manufacturer documentation, or medical director attestation . Features involving decentralized uploads, BYOD devices, scanner interoperability, optical character recognition (OCR), metadata mapping, AI-supported QC, and web-based reporting are classified as high-risk areas requiring formal validation. Validation protocols must outline learning objectives, acceptance criteria, sample size justification (often aligned with a 95% confidence interval), known controls, repeatability and reproducibility requirements, deviation handling, retest documentation, and medical director approval. These expectations are explicitly defined in DMS-108833 under “Protocol Deliverables” and form the basis for test execution and reporting .

All validation activities follow a structured lifecycle: Pre-Planning → Execution → Post-Validation → Launch, as shown in the Validation Logistics workflow (ECA-108646). Pre-planning includes creation of the Validation Plan, protocol templates, test environments, and Site URL configuration. Execution includes device testing (e.g., BYOD validation), lab testing, functional testing, and operational workflow testing. Post-validation includes routing completed test records for signatures, updating the Validation Summary Report (DMS-108566), and obtaining the U.S. Laboratory Medical Director’s DocuSign attestation, which is mandatory before patient testing can begin. After validation is complete and all signatures are collected, the feature or release may move to production and subsequently to customer deployment, supported by training, updated SOPs, and release notes .

All validation documentation—including Validation Plans, IQ/OQ/PQ protocols, executed test records, deviations, retest evidence, medical attestation, and summary reports—is retained within the Diagnexia IMS in compliance with the Document Control Procedure (DMS-8338). Any change requiring re-validation is managed through the Change Management Procedure (DMS-19364 and DCX-SOP-28), and new validation cycles are initiated when software features, hardware configurations, integrations, or workflows undergo modifications that may impact clinical validity, data integrity, regulatory compliance, or patient safety.

11. QUALIFICATION (IQ / OQ / PQ)

Installation Qualification (IQ) activities for the MyDiagnexia platform are performed to verify and document that all software, cloud infrastructure components, and Diagnexia-owned on-premise equipment are installed, configured, and deployed in accordance with approved design specifications, manufacturer requirements, and Diagnexia’s operational standards. The IQ process ensures that the validated environment supports secure, reliable, and compliant clinical telepathology workflows as required by CAP, CLIA, and ISO 15189.

Because MyDiagnexia is a fully cloud-based web platform, IQ is focused primarily on validating the U.S. AWS deployment, including provisioning of all functional application services, database instances, image rendering engines, storage buckets, QC Cloud infrastructure, API endpoints, user authentication services, logging mechanisms, encryption controls, and role-based access configurations. IQ confirms that all deployed cloud components match the approved software build,

URL endpoints, architecture diagrams, and configuration specifications referenced in the system design documentation and the Clinical Validation Plan (DMS-108833). This includes verification of build identifiers, version control lineage, environment isolation, and the operational readiness of all cloud-based modules supporting viewer rendering, case ingestion, QC workflows, reporting tools, and LIS/LIMS interoperability.

In alignment with the hybrid IQ model, Diagnexia also performs IQ on Diagnexia-owned scanners placed at customer sites, as these devices fall under Diagnexia's responsibility for commissioning, configuration, and regulatory compliance. Scanner IQ includes verification of installation location, unit serial numbers, firmware versions, configuration profiles, DICOM converter settings, calibration documentation, network connectivity, labeling requirements, and any scanner-specific integration points defined under DMS-69916 (Integration Configuration Work Instruction). This is consistent with the expectations outlined in the Quality Risk Assessment (DMS-105073), which identifies scanner interoperability, DICOM conversion, and image quality consistency as key risk domains requiring documented commissioning before use in digital workflows. Customer-owned scanners remain outside the scope of Diagnexia's IQ but are subject to compatibility testing and metadata validation during customer onboarding and User Acceptance Testing (UAT).

For cloud and on-prem components alike, all IQ activities include review of installation instructions, verification of correct component setup, confirmation of security configurations (e.g., encryption, SSO linking, MFA enablement when applicable), validation of audit log activation, storage retention configurations (e.g., WASABI or S3 settings), and assessment of network and performance prerequisites. Any deviations identified during IQ execution are documented and dispositioned per the Change Management Procedure (DCX-SOP-28), with corrective actions and retesting performed as required. IQ results, along with supporting evidence such as screenshots, installation records, configuration files, or network verification logs, are recorded in the executed IQ protocol and retained within the Diagnexia IMS in accordance with the Document Control Procedure (DMS-8338).

Scanner IQ, cloud IQ, and software IQ together form the foundation for subsequent Operational Qualification (OQ) and Performance Qualification (PQ) activities. Completion and approval of IQ are required before OQ and PQ testing can begin, as reflected in the Clinical Validation / Verification Plan (DMS-108833) and the structured validation lifecycle diagram contained in the Validation Logistics documentation. IQ completion must be approved by QA, Engineering, and the U.S. Laboratory Operations lead, and must be acknowledged by the U.S. Laboratory Medical Director in both the testing workflow and final Validation Summary Report (DMS-108566), ensuring the system is installed in a controlled, compliant, and documented manner prior to clinical validation or patient-care use.

12. CHANGE CONTROL

The goal of change control is to prevent “surprise” behavior in production by making sure every change is assessed, tested, documented, and approved before it goes live. All changes affecting the MyDiagnexia platform—including software enhancements, workflow adjustments, scanner integrations, configuration updates, or documentation modifications—are governed by the Diagnexia Change Management framework as defined in DMS-19364 (Procedure for Change Management) and executed through the Orcanos eQMS workflow described in DMS-80788 (Work Instruction for Change Management). These policies ensure that every change is evaluated, documented, controlled, and approved in a manner that maintains regulatory compliance, data integrity, patient safety, system stability, and operational readiness.

Under this framework, each change—such as those introduced in ECO-108614 for the MyDiagnexia Group Reporting Hub—is first classified as minor or major, based on its scope, impact, and

associated risk. ECO-108614 is designated as a *Major Change*, consistent with DMS-19364 criteria requiring multi-area impact assessment and full cross-functional review. The initial categorization triggers a structured impact assessment covering: QMS/SOP impacts, regulatory implications (CAP/CLIA/ISO 15189), information security (ISMS) impact, privacy implications requiring DPIA review, technical integration effects, operational workflow changes, and any diagnostic-specific impacts such as those affecting case routing, OCR workflows, or reporting logic. These assessments must be completed by the assigned SMEs prior to approval, as mandated by DMS-19364 sections 5.3 and 5.4 .

Where system functionality is affected, changes may require regression testing or re-validation, consistent with CAP requirements for software modification testing (GEN.43022) and the U.S. Clinical Validation Plan (DMS-108833). Regression testing ensures the change has not adversely impacted existing viewer behavior, QC Cloud performance, scanner interoperability, metadata mapping, or reporting workflows. Validation or verification activities are initiated as child actions linked directly to the ECO in Orcanos, following the structured routing and approval lifecycle defined in DMS-80788, where each task is assigned clear ownership, due dates, and evidence obligations .

Documentation updates—including SOP revisions, work instructions, training materials, and customer-facing guides—must be performed whenever the change affects how the system is used, validated, deployed, or maintained. These updates are tracked as ECO Action Items and must be completed, reviewed, and approved before an ECO can be closed. The Change Owner is responsible for coordinating these updates, while QA ensures compliance with the Document Control Procedure (DMS-8338).

Once all assessments, testing, documentation updates, and training actions are complete, the change is routed for formal approval in Orcanos. Required approvers may include QA, Regulatory Affairs, Information Security, the Data Protection Officer, the System Owner, and the Senior Director of QA/RA depending on risk level and impact, as defined in the approval matrix in DMS-19364 section 5.5 . After final approval, the change is implemented according to the validated plan, and release notes are generated to communicate the change to internal teams and customers as appropriate. For high-risk or clinically impactful changes, release notes must be accompanied by training verification and stakeholder communication records, per DMS-19364 and DMS-80788.

Through these controls, Diagnexia ensures that all MyDiagnexia platform changes—including those supporting Phase 1 digital uploads, group reporting, case assignment enhancements, and web-based reporting tools—are implemented in a structured, traceable, compliant, and safe manner consistent with CAP, CLIA, ISO 15189, HIPAA, and IMS requirements.

13. INCIDENT & DEFECT MANAGEMENT

Incident and defect management exists to ensure that when something goes wrong, we capture it, fix it, learn from it, and prevent it from happening again. Incident and defect management for the MyDiagnexia platform is governed by the Deciphex Procedure for Incident Management (DMS-8077, Rev C-2), which establishes a formal, structured, and auditable process for identifying, classifying, escalating, treating, and closing all incidents that may impact patient safety, product quality, data integrity, information security, or customer experience . This procedure applies to all Diagnexia systems and services—including cloud-based components, customer-validated configurations, and

any systems handling PHI or PII—and follows a shared model of responsibility across Deciphex Group entities.

Incidents may originate from multiple sources, including customer reports, internal QC checks, system monitoring alerts, audit findings, scanner workflow anomalies, OCR mismatches, metadata errors, access control failures, or security events such as phishing attempts or malware detection. Upon identification, the reporter must immediately complete DCX-FRM-05 Part A, triggering classification of the event as a *Customer Incident*, *Security Incident*, or *Data Breach*. The Data Protection Officer (DPO), Information Security team, Quality Assurance, and the Laboratory Director/Chief Medical Officer (CMO) jointly assess clinical impact, PHI/PII exposure, confidentiality/integrity/availability risk, and whether the incident requires regulatory notification under HIPAA, GDPR, or CAP/CLIA requirements .

All incidents are logged and evaluated using the workflow on page 4 of DMS-8077, which details the branching pathways for security incidents, data breaches, and customer-reported failures. Clinical incidents are assessed using DCX-FRM-05 Part B, and all incidents undergo a multi-disciplinary review to determine root cause, immediate containment, and whether corrective actions are required. If investigation reveals that a system change, software patch, or workflow modification is needed, a Change Request (ECO) is raised in accordance with the Change Management Procedure (DMS-19364), linking the incident to OQ/PQ re-testing, regression analysis, or documentation updates as required.

Defects arising from software behavior—such as viewer rendering errors, QC Cloud defects, metadata mismatches, pathologist assignment errors, or API failures—are documented in Jira and linked to customer or internal incidents per DMS-8077 Section 5.3. These defects are reviewed by Engineering, QA, and Information Security to determine whether they represent isolated functional bugs or broader systemic issues that may require CAPA initiation under DMS-8295 (Corrective Action & Improvement) or risk register updates (DCX-REG-00 or DCX-REG-19) .

Incident treatment and resolution follow DMS-8077 Section 5.5, requiring immediate containment, identification of temporary and permanent countermeasures, and root-cause remediation. All decisions, actions, and communications—including any customer notifications, CAP/CLIA-required actions, HIPAA reporting requirements, or data-breach notifications—are documented in DCX-FRM-05 Part C and stored in the Security & Data Breach Incident Database (DCX-REG-19) for full traceability. Records are retained to ensure audit readiness, regulatory defensibility, and forensic continuity.

Incidents demonstrating recurrence trends are reviewed periodically to ensure that prior countermeasures have been effective; if patterns emerge, top management may be engaged and additional corrective action or external expertise may be required. Through this structured process, MyDiagnexia ensures that all incidents—clinical, security, operational, or customer-reported—are managed consistently, promptly, and in alignment with regulatory expectations for digital pathology systems and cloud-based telepathology workflows.

14. TRAINING & COMPETENCY

Training must follow DMS-8082. Training and competency for MyDiagnexia follow the requirements of the Diagnexia Training Procedure (DMS-8082) and associated competency frameworks that govern digital pathology operations, personnel qualification, and clinical use of telepathology systems. All individuals involved in the operation, validation, support, or clinical use of MyDiagnexia—including pathologists, laboratory staff, operational personnel, engineering, customer-facing teams, and

quality/regulatory staff—must complete role-specific training prior to performing any activities that could impact patient safety, product quality, data integrity, or system configuration. This requirement is consistent with CLIA personnel standards and CAP GEN.51728, which mandates documented training for all telepathology users, as well as GEN.52900, which requires training for all individuals using whole slide imaging systems in clinical workflows.

Training requirements are defined during onboarding, system updates, ECO-driven changes, and prior to rollout of new features. As outlined in the U.S. Clinical Validation / Verification Plan (DMS-108833), personnel executing validation protocols must be trained and qualified using competency-based materials, including DMS-41059 for QC of digital images and DMS-8354 for general personnel competency assessments. Any training dependencies identified during impact assessments for ECO-108614 (Phase 1 rollout) must be completed before the change is released to production, and validation testers must demonstrate documented proficiency before participating in IQ/OQ/PQ testing.

Given the cloud-based nature of MyDiagnexia and the transition to BYOD for pathologists, training includes mandatory reading and attestation to the BYOD device validation requirements (DMS-94327), which cover minimum screen specifications, security expectations, browser requirements, and user responsibilities related to HIPAA and PHI access. Pathologists must attest that they understand and accept their obligations under Diagnexia's privacy, security, and telepathology workflows, and all BYOD devices must be validated prior to clinical use. This aligns with risk controls identified in the Quality Risk Assessment (DMS-105073), which require technical competency and device readiness verification before cloud-based reporting can proceed.

All training activities—including training related to QC Cloud workflows, image uploading, scanner interactions (for Diagnexia-owned scanners), case assignment features, reporting tools, metadata correction processes, OCR fallback workflows, LIS/LIMS integration behavior, and Incident Management procedures—must be documented in the appropriate system of record (BambooHR for personnel files or Orcanos when linked to an ECO). Competency assessments may include proficiency testing, read-and-understand confirmation, observed workflow demonstrations, scenario-based exercises, or completion of validation-related tasks, depending on the role and training content. For pathologists performing clinical sign-out, additional training may be required for screen calibration, viewer tools, and reporting templates.

When incidents or corrective actions occur, DMS-8077 requires reassessment of training needs. If an incident analysis identifies training gaps or user error as a contributing factor, new training or retraining must be assigned, documented, and included as part of the corrective action plan. Updated SOPs, work instructions, or system changes introduced through ECOs—such as ECO-108614 for Phase 1—trigger automatic training review to ensure that all end users understand the updated workflows prior to the change going live.

Verification of training and competency must occur before the platform is released for patient use and before any user interacts with the clinical environment. The U.S. Laboratory Medical Director must confirm that all personnel supporting clinical workflows have completed required training as part of the validation summary (DMS-108566). Through these controls, Diagnexia ensures that all MyDiagnexia users are properly trained, competent, and qualified to operate the system safely and in accordance with CAP, CLIA, ISO 15189, HIPAA, and Diagnexia IMS requirements.

Required training:

- **Viewer usage**

- **Web reporting**
- **QC Cloud workflows**
- **LIS integration**
- **BYOD device validation (DMS-94327)**
- **Privacy, security, HIPAA**
- **Clinical workflow training**
- **U.S. procedures (slide receipt, QC, turnaround time, error handling)**

Training records stored in:

- **BambooHR**
 - **Orcanos (where applicable)**
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15. LIFECYCLE DOCUMENTATION

Documentation and records management for the MyDiagnexia platform are governed by the Deciphex Procedure for Document Control (DMS-8338) and the corresponding records management expectations within the Integrated Management System (IMS). All controlled documents—including Quality Plans, SOPs, work instructions, user guides, release notes, and technical specifications—must be authored, reviewed, approved, version-controlled, and archived in accordance with DMS-8338 to ensure they remain current, accurate, and aligned with the validated state of the system.

In accordance with the Diagnexia U.S. Clinical Validation / Verification Plan (DMS-108833), all validation documents are created, approved, executed, and stored exclusively in Orcanos, which serves as the system of record for validation lifecycle documentation. This includes Validation Plans, IQ/OQ/PQ protocols, executed test cases, test evidence, screenshots, deviations, retest documentation, traceability matrices (RTMs), validation summary reports (DMS-108566), medical director attestations, and approval routing records. The use of Orcanos ensures full traceability, controlled revision history, secure e-signature workflows, and audit readiness consistent with CAP, CLIA, ISO 15189, and 21 CFR Part 11-compliant signature capture where required.

Records produced outside formal validation activities—such as ECO documentation, impact assessments, risk assessments, DPIAs, scanner commissioning records, BYOD validation records (DMS-94327), incident records, and CAPA documentation—must also be uploaded into Orcanos when they support a change, corrective action, or validation deliverable. Operational records generated through the Incident Management Procedure (DMS-8077), including DCX-FRM-05 forms (Parts A/B/C), DCX-REG-19 entries, and security incident evidence, are retained in Orcanos when associated with a corrective action or change control, and maintained in accordance with their respective retention schedules.

For broader IMS compliance, additional non-validation records (e.g., training records, SOP updates, versioned user guides, personnel competency assessments, board meeting minutes) are stored in their designated systems of record—primarily BambooHR for personnel training and the IMS Document Control Library for SOPs and policies—while still being referenced within Orcanos when linked to an ECO or validation activity. This ensures end-to-end traceability between training, system changes, and validation outcomes, consistent with CAP GEN.51728 and CLIA personnel competency expectations.

Retained documentation must satisfy all applicable regulatory retention requirements, including CAP/CLIA obligations for telepathology, data integrity, WSI system validation, system change verification, and audit trail preservation. All records must be complete, contemporaneous, legible, attributable, and securely stored throughout their retention period, and must be readily retrievable for internal audits, CAP inspections, CLIA surveys, customer audits, regulatory review, or forensic investigations.

Through this framework—rooted in DMS-8338, DMS-108833, and Orcanos-based record retention—Diagnexia ensures that all documentation and records associated with the MyDiagnexia platform are controlled, traceable, compliant, and reflective of the validated state of the system throughout its lifecycle.

16. SECURITY & PRIVACY

Compliance with:

- HIPAA
- ISO 27001
- Diagnexia Security Policies
- DPIA requirements

Security and privacy controls for MyDiagnexia follow Diagnexia's ISO 27001-aligned Information Security Management System (ISMS) and are governed by the Master Security Document (DMS-72576), Data Protection Impact Assessment (DPIA) requirements (DMS-8819), the Privacy Policy (DMS-8060 / DMS-106470), and the Deciphex Information Security policies referenced throughout the IMS. MyDiagnexia handles Protected Health Information (PHI) and Personally Identifiable Information (PII); therefore, all system components must comply with HIPAA, relevant state privacy laws, contractual Business Associate Agreement (BAA) requirements, and GDPR/PIPEDA where applicable.

Security measures include mandatory encryption in transit and at rest, secure AWS deployment within Diagnexia-controlled regions, role-based access control (RBAC), audit logging, intrusion detection, and continuous monitoring. Authentication must follow Diagnexia's approved access-management controls, and SSO/MFA must be enabled where required by customer policies or IS risk assessments. As described in the Quality Risk Assessment (DMS-105073), decentralized pathologist access introduces specific security requirements; thus, pathologists must meet BYOD security criteria (DMS-94327), including full-disk encryption, supported browser versions, secure session handling, and minimum screen specifications.

Any changes to security-relevant components must undergo ISMS review as defined in DMS-19364 and DMS-80788, including impact assessments across confidentiality, integrity, and availability. All PHI/PII incidents must be evaluated and managed according to DMS-8077 (Incident Management) and DMS-8083 (Personal Data Breach Procedure), with involvement from the DPO, Information Security, and the Laboratory Director where clinical impact may exist. Regulatory notifications (e.g., HIPAA Breach Notification Rule) are executed only under the authority of the DPO.

Security and privacy documentation—including DPIAs, access reviews, audit logs, incident response forms, and ISMS risk register entries—must be retained per the Document Control Procedure (DMS-8338) and be readily available for CAP, CLIA, or regulatory inspection. Through this ISMS framework, MyDiagnexia ensures the confidentiality, integrity, and availability of PHI while supporting compliant, secure clinical workflows.

Security controls include:

- Audit logging
- Access control
- MFA (if enabled)
- Secure cloud storage
- Encrypted transmission
- Role-based access controls

17. LIS/LIMS INTEGRATION COMPLIANCE

Integrations must follow DMS-69916 including:

- HL7 ORM/ORU validation
- Message traceability
- Patient metadata accuracy
- Round-trip testing
- Backup workflows

MyDiagnexia supports interoperability with laboratory information systems (LIS/LIMS) using HL7 v2.x, FHIR-based APIs, and secure cloud-based data exchange. All integrations must follow DMS-69916 (Integration Configuration Work Instruction), which defines configuration, mapping, testing, and verification requirements. LIS testing must comply with CAP GEN.43022, which requires validation of any new or modified LIS or telepathology interface prior to clinical use.

Each integration undergoes configuration validation to confirm correct message mapping (ORM/ORU), correct case-ID/slide-ID pairing, metadata accuracy, receipt acknowledgment, and failover behavior. OCR fallback and metadata correction workflows—identified as moderate-to-high risk in the QRA—must be validated during integration testing. All integration-related changes must be documented as ECO action items and undergo appropriate OQ/PQ testing before customer go-live. Any customer-specific workflows that interface with MyDiagnexia must be validated by the customer's laboratory in accordance with CLIA and CAP requirements, with Diagnexia providing integration testing support and technical documentation.

18. CUSTOMER VALIDATION

Each customer must execute UAT:

- **LIS connectivity**
- **Workflow simulation**
- **Reporting validation**
- **Credentialing**
- **Slide upload tests**
- **Turnaround time validation**

Customer validation is required for any clinical customer adopting MyDiagnexia. While Diagnexia performs system-level validation (IQ/OQ/PQ), each customer must complete validation of their local workflow, in alignment with CAP and CLIA expectations for telepathology systems (GEN.40630, GEN.50630) and LIS interfaces (GEN.43022). Customer validation includes workflow simulation, LIS integration testing (if applicable), verification of slide upload pathways, reporting workflows, and case assignment rules.

Diagnexia provides customer-specific validation templates (e.g., UAT scripts, checklist-based validation), while customers are responsible for verifying accuracy, completeness, and intended-use suitability within their environment. The Diagnexia implementation team supports customer validation by providing test data, test instructions, scanner configuration files (where Diagnexia owns the scanner), and training materials. Evidence of successful customer validation should be retained by the customer and referenced when necessary by Diagnexia's customer success teams.

19. RELEASE MANAGEMENT

Release Management follows Diagnexia's Secure SDLC (DMS-8333), Release Procedures (DMS-20398), and the Change Management Process (DMS-19364, DMS-80788). Before any MyDiagnexia release is promoted to production, it must have completed all required IQ/OQ/PQ validation activities with documentation stored in Orcanos. Release packages must include release notes, known issues, updated user documentation, and evidence of all required approvals.

Major changes—such as those defined in ECO-108614—must undergo multi-departmental review, including QA, Engineering, Regulatory, Information Security, and the DPO when PHI or PII is impacted. The U.S. Laboratory Director or designated Medical Director must sign the Validation Summary Report (DMS-108566) prior to clinical release. Release notes must be communicated to internal stakeholders and customers (where applicable), and training must be completed prior to enabling new features.

Rollback plans must be developed and assessed for all high-risk releases, as mandated in DMS-19364. All production releases must be fully traceable, documented in Orcanos, and available for regulatory inspection.

Release package includes:

- Release Notes
- Known Issues
- VSR
- Training materials
- Updated SOPs

20. POST-IMPLEMENTATION MONITORING

KPIs:

- Upload success rate
- Viewer performance
- Pathologist turnaround time
- Pending consult metrics
- QC defect rate
- Slide mapping issues
- Integration message failures

Post-implementation monitoring ensures the continued performance, reliability, and compliance of MyDiagnexia after release. Monitoring includes viewer performance metrics, QC Cloud processing success rates, upload error frequencies, OCR mismatch rates, case-assignment behavior, integration message failures, and pathologist BYOD performance. These metrics are reviewed periodically by Engineering, Clinical Operations, IS, and QA to detect trends that may require corrective action or system enhancement.

Customer feedback (tickets, incident reports, usability feedback, training gaps) is monitored through the Customer Support system and linked to incidents (DMS-8077) or CAPA processes (DMS-8295) where required. Post-deployment monitoring also includes validation of audit logging, access control events, cloud performance, and multi-site reporting behavior. Recurring issues must be escalated and addressed per the Corrective Action Procedure.

21. RETIREMENT

System retirement must ensure:

- **Data migration**
- **Data retention (CLIA: 10+ years for digital images depending on state)**
- **Secure decommissioning**
- **Customer notification**

System retirement for MyDiagnexia follows the principles of DMS-8338 (Document Control), DMS-8079 (Record Management), and applicable CAP/CLIA retention requirements. Retirement activities require formal change control (DMS-19364) and include archival of all documentation in Orcanos, secure migration or retention of PHI/PII in accordance with laboratory retention laws (typically 10+ years for WSI records), decommissioning of technical components, validation of data integrity post-archive, and customer notifications.

A controlled retirement plan must address cloud service shutdown procedures, access removal, data export options, contract closure, and post-retirement verification. Security controls remain in effect until full decommissioning is complete, ensuring confidentiality, integrity, and availability of all preserved records.

22. APPENDICES

Appendix A — Document List

Appendix B — Validation Deliverables

Appendix C — Risk Register Summary

Appendix D — Roles & Contact List

Appendix E — Training Matrix

Appendix F — Glossary

Appendix G — State Licensure Considerations

Appendix H — BYOD Device Minimum Specs

23. REVIEW AND APPROVAL

This document shall be reviewed annually or as needed due to changes in scope, regulatory requirements, or release schedule.